

Assignment No. 6

Hub and Spoke Architecture using AWS Transit Gateway

Objective:

The objective of this assignment is to design and implement a scalable **Hub and Spoke network architecture** in AWS using **AWS Transit Gateway**. Multiple Virtual Private Clouds (VPCs) are interconnected through a central Transit Gateway to enable secure communication while simplifying network management.

Problem Statement:

Organizations often manage multiple VPCs for different departments or applications. Establishing individual VPC peering connections between every VPC becomes difficult as the number of VPCs increases.

AWS Transit Gateway solves this issue by acting as a central routing hub that connects multiple VPCs together.

AWS Services Used

- Amazon VPC
- Amazon EC2
- AWS Transit Gateway
- Transit Gateway Attachments
- Internet Gateway
- Route Tables
- Security Groups
- Network ACL
- Public Subnets
- Elastic Network Interface

Network Design:

Hub VPC

Property	Value
CIDR Block	10.1.0.0/16
Public Subnet	10.1.1.0/24
EC2 Private IP	10.1.1.236

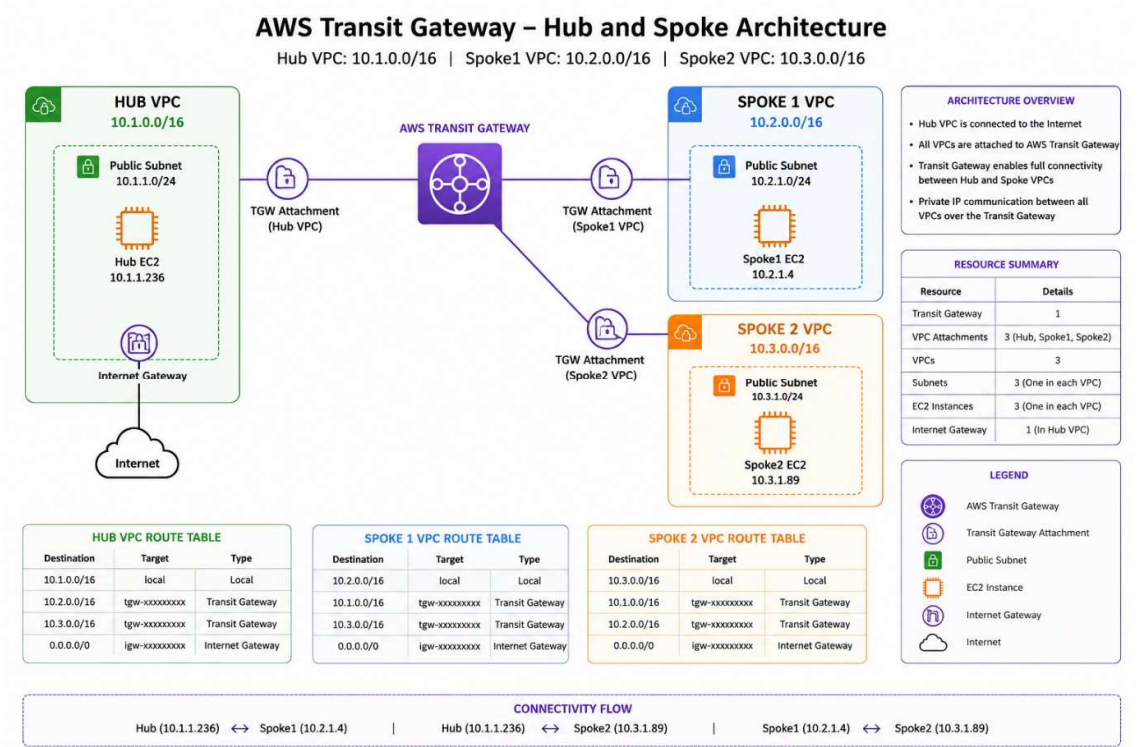
Spoke VPC 1

Property	Value
CIDR Block	10.2.0.0/16
Public Subnet	10.2.1.0/24
EC2 Private IP	10.2.1.4

Spoke VPC 2

Property	Value
CIDR Block	10.3.0.0/16
Public Subnet	10.3.1.0/24
EC2 Private IP	10.3.1.89

Architecture:



Implementation Steps

Step 1: Create Hub VPC

A Hub VPC was created with CIDR block **10.1.0.0/16**. This VPC acts as the central communication point for all spoke VPCs.

Your VPCs (1/2) Info

Last updated less than a minute ago

Actions

Create VPC

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
-	vpc-055fefadc90488fe4	Available	-	-	Off	172
hub-vpc	vpc-0f0efdfcf5b8462be	Available	-	-	Off	10.

vpc-0f0efdfcf5b8462be / hub-vpc

Details

VPC ID

vpc-0f0efdfcf5b8462be

DNS resolution

Enabled

Main network ACL

-

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-0cb767041dfbe93a1

IPv4 CIDR

10.1.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

-

IPv6 pool

-

Owner ID

303602054364

Step 2: Create Spoke VPC 1

A second VPC named **Spoke VPC 1** was created with CIDR block **10.2.0.0/16**.

Your VPCs (1/3) Info

Last updated less than a minute ago

Actions

Create VPC

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
-	vpc-055fefadc90488fe4	Available	-	-	Off	172
hub-vpc	vpc-0f0efdfcf5b8462be	Available	-	-	Off	10.
spoke-vpc-1	vpc-0db8a83885413c74d	Available	-	-	Off	10.

vpc-0db8a83885413c74d / spoke-vpc-1

Details

VPC ID

vpc-0db8a83885413c74d

DNS resolution

Enabled

Main network ACL

-

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-0cb767041dfbe93a1

IPv4 CIDR

10.2.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

-

IPv6 pool

-

Owner ID

303602054364

Step 3: Create Spoke VPC 2

A third VPC named **Spoke VPC 2** was created with CIDR block **10.3.0.0/16**.

Your VPCs (1/4) Info

Last updated less than a minute ago

Actions

Create VPC

Find VPCs by attribute or tag

< 1 >

	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
☐	hub-vpc	vpc-0f0efdfcf5b8462be	Available	-	-	Off	10...
☐	spoke-vpc-1	vpc-0db8a83885413c74d	Available	-	-	Off	10...
☒	spoke-vpc-2	vpc-04bf643d7c728a5df	Available	-	-	Off	10...

vpc-04bf643d7c728a5df / spoke-vpc-2

Details

VPC ID

vpc-04bf643d7c728a5df

DNS resolution

Enabled

Main network ACL

-

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-0cb767041dfbe93a1

IPv4 CIDR

10.3.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

-

IPv6 pool

-

Owner ID

303602054364

Step 4: Create Subnets

One subnet was created inside each VPC.

Hub Subnet

CIDR:

10.1.1.0/24

You have successfully created 1 subnet: subnet-02578890ce3a03ad3

Subnets (1/1) Info

Last updated less than a minute ago

Actions

Create subnet

Find subnets by attribute or tag

< 1 >

Subnet ID : subnet-02578890ce3a03ad3

Clear filters

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☒	hub-subnet	subnet-02578890ce3a03ad3	Available	vpc-0f0efdfcf5b8462be hub-vpc	Off	10.1.1.0/24

subnet-02578890ce3a03ad3 / hub-subnet

Subnet ID

subnet-02578890ce3a03ad3

IPv4 CIDR

10.1.1.0/24

Availability Zone

aps1-south-1 (ap-south-1a)

Network ACL

-

Auto-assign customer-owned IPv4 address

No

Subnet ARN

arn:aws:ec2:ap-south-1:303602054364:subnet/subnet-02578890ce3a03ad3

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

State

Available

IPv6 CIDR

-

VPC

vpc-0f0efdfcf5b8462be | hub-vpc

Auto-assign public IPv4 address

No

Outpost ID

-

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-0453771794e9bbeb3

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Spoke 1 Subnet

CIDR:
10.2.1.0/24

You have successfully created 1 subnet: subnet-066edba299fa3259c

Subnets (1/1) info

Last updated less than a minute ago

Actions

Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-066edba299fa3259c

Clear filters

☒	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☒	spoke1-subnet	subnet-066edba299fa3259c	Available	vpc-0db8a83885413c74d spo...	Off	10.2.1.0/24

subnet-066edba299fa3259c / spoke1-subnet

Subnet ID

subnet-066edba299fa3259c

IPv4 CIDR

10.2.1.0/24

Availability Zone

aps1-az1 (ap-south-1a)

Network ACL

-

Auto-assign customer-owned IPv4 address

No

Subnet ARN

arn:aws:ec2:ap-south-1:303602054364:subnet/subnet-066edba299fa3259c

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

State

Available

IPv6 CIDR

-

VPC

vpc-0db8a83885413c74d | spoke-vpc-1

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

-

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-0d9fe589ad8252603

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

-

Spoke 2 Subnet

CIDR:
10.3.1.0/24

You have successfully created 1 subnet: subnet-0de0e2c6b04a8962d

Subnets (1/6) info

Last updated less than a minute ago

Actions

Create subnet

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	hub-subnet	subnet-02578890ce3a03ad3	Available	vpc-0f0efdfcf5b8462be hub-vpc	Off	10.1.1.0/24
☐	spoke1-subnet	subnet-066edba299fa3259c	Available	vpc-0db8a83885413c74d spo...	Off	10.2.1.0/24
☒	spoke2-subnet	subnet-0de0e2c6b04a8962d	Available	vpc-04bf643d7c728a5df spok...	Off	10.3.1.0/24

subnet-0de0e2c6b04a8962d / spoke2-subnet

Subnet ID

subnet-0de0e2c6b04a8962d

IPv4 CIDR

10.3.1.0/24

Availability Zone

aps1-az1 (ap-south-1a)

Network ACL

-

Auto-assign customer-owned IPv4 address

No

Subnet ARN

arn:aws:ec2:ap-south-1:303602054364:subnet/subnet-0de0e2c6b04a8962d

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

State

Available

IPv6 CIDR

-

VPC

vpc-04bf643d7c728a5df | spoke-vpc-2

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

-

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-04c4112b2f8827cf1

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

-

Step 5: Attach Internet Gateway

An Internet Gateway was created and attached to the Hub VPC to provide internet connectivity.

Internet gateways (1/2) Info

Find internet gateways by attribute or tag

	Name	Internet gateway ID	State	VPC ID	Owner
☐	-	igw-060dbee86a30857e7	Attached	vpc-055fefadc90488fe4	303602054364
☒	hub-igw	igw-0e879ed8dcace836a	Attached	vpc-0f0efdfcf5b8462be hub-vpc	303602054364

igw-0e879ed8dcace836a / hub-igw

Details

Tags

Details

Internet gateway ID

igw-0e879ed8dcace836a

State

Attached

VPC ID

vpc-0f0efdfcf5b8462be | hub-vpc

Owner

303602054364

Step 6: Configure Initial Hub Route Table

The Hub Route Table was configured with:

- Local Route
- Default Route (0.0.0.0/0) pointing to Internet Gateway

Updated routes for rtb-0453771794e9bbeb3 successfully

Details

rtb-0453771794e9bbeb3

Actions

Details Info

Route table ID

rtb-0453771794e9bbeb3

VPC

vpc-0f0efdfcf5b8462be | hub-vpc

Main

Yes

Owner ID

303602054364

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Filter routes

Both Edit routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0e879ed8dcace836a	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table

Step 7: Create AWS Transit Gateway

An AWS Transit Gateway was created to connect all VPCs through a centralized routing architecture.

You successfully created tgw-0bebd2ee4e0f72b3e / assignment6-tgw.

You can visualize and monitor your Transit Gateway(s) from the AWS Network Manager. Register your Transit Gateway by creating a global network to get started.

Transit gateways (1/1) Info

Find transit gateway by attribute or tag

1

Name

assignment6-tgw

Transit gateway ID

tgw-0bebd2ee4e0f72b3e

Owner ID

303602054364

State

Available

Transit gateway: tgw-0bebd2ee4e0f72b3e / assignment6-tgw

Transit gateway ID

tgw-0bebd2ee4e0f72b3e

Transit gateway ARN

arn:aws:ec2:ap-south-1:303602054364:transit-gateway/tgw-0bebd2ee4e0f72b3e

Owner ID

303602054364

Description

Hub and Spoke Architecture

State

Available

Default association route table

Enable

Default propagation route table

Enable

Transit gateway CIDR blocks

-

Amazon ASN

64512

Association route table ID

tgw-rtb-0e4750170df2a9c01

Propagation route table ID

tgw-rtb-0e4750170df2a9c01

Security Group Referencing support

Disable

DNS support

Enable

Auto accept shared attachments

Disable

VPN ECMP support

Enable

Metering policy ID

-

Step 8: Create Transit Gateway Attachments

Each VPC was attached to the Transit Gateway.

Attachments created:

- Hub Attachment
- Spoke 1 Attachment
- Spoke 2 Attachment

You can visualize and monitor your Transit Gateway(s) from the AWS Network Manager. Register your Transit Gateway by creating a global network to get started.

Transit gateway attachments (3) Info

Find transit gateway attachment by attribute or tag

1

Name

spoke1-attachment

Transit gateway attachment ID

tgw-attach-03d1f4fcb1ddda492

Transit gateway ID

tgw-0bebd2ee4e0f72b3e

State

Available

Resource type

VPC

Resource ID

vpc-0db8a83885413c74d

spoke2-attachment

tgw-attach-0a43452152fa53c64

tgw-0bebd2ee4e0f72b3e

Available

VPC

vpc-04bf643d7c728a5df

hub-attachment

tgw-attach-0ac71192e0ac39d68

tgw-0bebd2ee4e0f72b3e

Available

VPC

vpc-0f0e9dfcf5b8462be

Select a transit gateway attachment

Step 9: Configure VPC Route Tables

After attaching the Transit Gateway, the route tables were updated.

Hub Route Table

Added Routes:

- 10.2.0.0/16 → Transit Gateway
- 10.3.0.0/16 → Transit Gateway

Route tables (1/4) Info

Last updated 16 minutes ago Actions Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-03bf17e8c64d0bb29	-	-	Yes	vpc-055fefadc90488fe4
-	rtb-0d9fe589ad8252603	-	-	Yes	vpc-0db8a83885413c74d sp-
-	rtb-0453771794e9bbeb3	-	-	Yes	vpc-0f0efdfcf5b8462be hub-
-	rtb-04c4112b2f8827cf1	-	-	Yes	vpc-04bf643d7c728a5df spo

rtb-0453771794e9bbeb3 Settings Dropdown

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (4)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0e879ed8dcace836a	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table
10.2.0.0/16	tgw-0bebd2ee4e0f72b3e	Active	No	Create Route
10.3.0.0/16	tgw-0bebd2ee4e0f72b3e	Active	No	Create Route

Spoke 1 Route Table

Added Routes:

- 10.1.0.0/16 → Transit Gateway
- 10.3.0.0/16 → Transit Gateway

Route tables (1/4) Info

Last updated 16 minutes ago

Actions

Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-03bf17e8c64d0bb29	-	-	Yes	vpc-055fefadc90488fe4
☒	rtb-0d9fe589ad8252603	-	-	Yes	vpc-0db8a83885413c74d spc
-	rtb-0453771794e9bbeb3	-	-	Yes	vpc-0f0efdfcf5b8462be hub-
-	rtb-04c4112b2f8827cf1	-	-	Yes	vpc-04bf643d7c728a5df spol

rtb-0d9fe589ad8252603

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (3)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.1.0.0/16	tgw-0bebd2ee4e0f72b3e	Active	No	Create Route
10.2.0.0/16	local	Active	No	Create Route Table
10.3.0.0/16	tgw-0bebd2ee4e0f72b3e	Active	No	Create Route

Spoke 2 Route Table

Added Routes:

- 10.1.0.0/16 → Transit Gateway
- 10.2.0.0/16 → Transit Gateway

Route tables (1/4) Info

Last updated 17 minutes ago

Actions

Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-03bf17e8c64d0bb29	-	-	Yes	vpc-055fefadc90488fe4
-	rtb-0d9fe589ad8252603	-	-	Yes	vpc-0db8a83885413c74d spc
-	rtb-0453771794e9bbeb3	-	-	Yes	vpc-0f0efdfcf5b8462be hub-
☒	rtb-04c4112b2f8827cf1	-	-	Yes	vpc-04bf643d7c728a5df spol

rtb-04c4112b2f8827cf1

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (3)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.1.0.0/16	tgw-0bebd2ee4e0f72b3e	Active	No	Create Route
10.2.0.0/16	tgw-0bebd2ee4e0f72b3e	Active	No	Create Route
10.3.0.0/16	local	Active	No	Create Route Table

Step 10: Launch EC2 Instances

One EC2 instance was launched in each VPC.

Hub EC2

Private IP:

10.1.1.236

Instances (1/3) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☐	spoke1-ec2	i-080ceb3b7c99d8618	Running	t3.micro	3/3 checks passed	ap-south-1a	-	1
☐	spoke2-ec2	i-0e75b25d631d04500	Running	t3.micro	3/3 checks passed	ap-south-1a	-	6
☒	hub-ec2	i-036680a1b73d1a20b	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-127-4-122.ap-s...	1

i-036680a1b73d1a20b (hub-ec2)

Instance ID

i-036680a1b73d1a20b

IPv6 address

-

Hostname type

IP name: ip-10-1-1-236.ap-south-1.compute.internal

Answer private resource DNS name

-

Public IPv4 address

13.127.4.122 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-1-1-236.ap-south-1.compute.internal

Instance type

t3.micro

Private IPv4 addresses

10.1.1.236

Public DNS

ec2-13-127-4-122.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses

-

Spoke 1 EC2

Private IP:

10.2.1.4

Instances (1/3) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☐	hub-ec2	i-0b383577c4c253ce1	Running	t3.micro	3/3 checks passed	ap-south-1a	-	3
☒	spoke1-ec2	i-080ceb3b7c99d8618	Running	t3.micro	3/3 checks passed	ap-south-1a	-	1
☐	spoke2-ec2	i-03c28cf371d28238a	Running	t3.micro	Initializing	ap-south-1a	-	1

i-080ceb3b7c99d8618 (spoke1-ec2)

Instance ID

i-080ceb3b7c99d8618

IPv6 address

-

Hostname type

IP name: ip-10-2-1-4.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

13.126.43.150 [Public IP]

Public IPv4 address

13.126.43.150 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-2-1-4.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0db8a83885413c74d (spoke-vpc-1)

Private IPv4 addresses

10.2.1.4

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations.

Spoke 2 EC2

Private IP:

10.3.1.89

Instances (1/4) info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
spoke2-ec2	i-0e75b25d631d04500	Running	t3.micro	3/3 checks passed	ap-south-1a	-

i-0e75b25d631d04500 (spoke2-ec2)

Instance ID

i-0e75b25d631d04500

IPv6 address

-

Hostname type

IP name: ip-10-3-1-89.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

65.2.171.153 [Public IP]

Public IPv4 address

65.2.171.153 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-3-1-89.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-04bf643d7c728a5df (spoke-vpc-2)

Private IPv4 addresses

10.3.1.89

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. [Learn more](#)

Step 11: Verify Transit Gateway Route Table

The Transit Gateway Route Table was checked to verify that all VPC attachments were correctly associated and propagated.

Associations

Verified that:

- Hub Attachment
- Spoke 1 Attachment
- Spoke 2 Attachment

were successfully associated.

Transit gateway route tables (1/1) Info

Find transit gateway route table by attribute or tag

Name

Transit gateway route table ID

Transit gateway ID

State

Default association route table

Default

tgw-rtb-0e4750170df2a9c01

tgw-0bebd2ee4e0f72b3e

Available

Yes

Yes

Transit gateway route tables: tgw-rtb-0e4750170df2a9c01

Details

Associations

Propagations

Prefix list references

Routes

Tags

Associations (3) Info

Find association by attribute or tag

Delete association

Create association

Attachment ID

Resource type

Resource ID

State

tgw-attach-03d1f4fcb1ddda492

VPC

vpc-0db8a83885413c74d

Associated

tgw-attach-0ac71192e0ac39d68

VPC

vpc-0f0efdfcf5b8462be

Associated

tgw-attach-0a43452152fa53c64

VPC

vpc-04bf643d7c728a5df

Associated

Propagations

Verified that all VPC routes were propagated successfully into the Transit Gateway Route Table.

Transit gateway route tables (1/1) Info

Find transit gateway route table by attribute or tag

Actions

Create transit gateway route table

☒	Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default
☒		tgw-rtb-0e4750170df2a9c01	tgw-0bebd2ee4e0f72b3e	Available	Yes	Yes

Transit gateway route tables: tgw-rtb-0e4750170df2a9c01

0.0.0.0/0

:::0

Routes (3) Info

Find route by attribute or tag

Actions

Create static route

☐	CIDR	Attachment ID	Resource ID	Resource type	Route type	Route state
☐	10.1.0.0/16	tgw-attach-0ac71192e0ac39d68	vpc-0f0e9dfcf5b8462be	VPC	Propagated	Active
☐	10.2.0.0/16	tgw-attach-03d1f4fcb1ddda492	vpc-0db8a83885413c74d	VPC	Propagated	Active
☐	10.3.0.0/16	tgw-attach-0a43452152fa53c64	vpc-04bf643d7c728a5df	VPC	Propagated	Active

Step 12: SSH Connection

SSH access to the Hub EC2 instance was successfully established using the key pair.

```
PS C:\Users\YASH\Downloads> ssh -i aws-keypair.pem ec2-user@13.127.4.122
#_
~\_  ####_      Amazon Linux 2023
~~ \_#####\
~~  \####|
~~   \#/  ---  https://aws.amazon.com/linux/amazon-linux-2023
~~    V~'  '--->
      ~~~
      ~~-.-.
      ~-/_  -/_
      ~/_m/'

Last login: Wed Jul  8 16:50:46 2026 from 112.110.121.20
[ec2-user@ip-10-1-1-236 ~]$ |
```

Step 13: Connectivity Test

Connectivity between Hub and Spoke VPCs was tested using the **ping** command.

Results:

- Hub → Spoke 1: Successful
- Hub → Spoke 2: Successful

This confirms that the Transit Gateway and route tables were configured correctly.

```
[ec2-user@ip-10-1-1-236 ~]$ ping -c 4 10.2.1.4
PING 10.2.1.4 (10.2.1.4) 56(84) bytes of data.
64 bytes from 10.2.1.4: icmp_seq=1 ttl=126 time=4.10 ms
64 bytes from 10.2.1.4: icmp_seq=2 ttl=126 time=0.540 ms
64 bytes from 10.2.1.4: icmp_seq=3 ttl=126 time=0.744 ms
64 bytes from 10.2.1.4: icmp_seq=4 ttl=126 time=0.530 ms

--- 10.2.1.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3089ms
rtt min/avg/max/mdev = 0.530/1.477/4.095/1.513 ms
[ec2-user@ip-10-1-1-236 ~]$ ping -c 4 10.3.1.89
PING 10.3.1.89 (10.3.1.89) 56(84) bytes of data.
64 bytes from 10.3.1.89: icmp_seq=1 ttl=126 time=3.10 ms
64 bytes from 10.3.1.89: icmp_seq=2 ttl=126 time=0.508 ms
64 bytes from 10.3.1.89: icmp_seq=3 ttl=126 time=0.537 ms
64 bytes from 10.3.1.89: icmp_seq=4 ttl=126 time=0.513 ms

--- 10.3.1.89 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3096ms
rtt min/avg/max/mdev = 0.508/1.163/3.096/1.115 ms
[ec2-user@ip-10-1-1-236 ~]$ |
```

Conclusion

This assignment successfully demonstrated the implementation of a Hub and Spoke Architecture using AWS Transit Gateway. Multiple VPCs were connected through a centralized Transit Gateway, route tables were configured correctly, and successful connectivity between all VPCs was verified using EC2 instances. The implementation shows how Transit Gateway simplifies inter-VPC communication and improves network scalability and management in AWS.